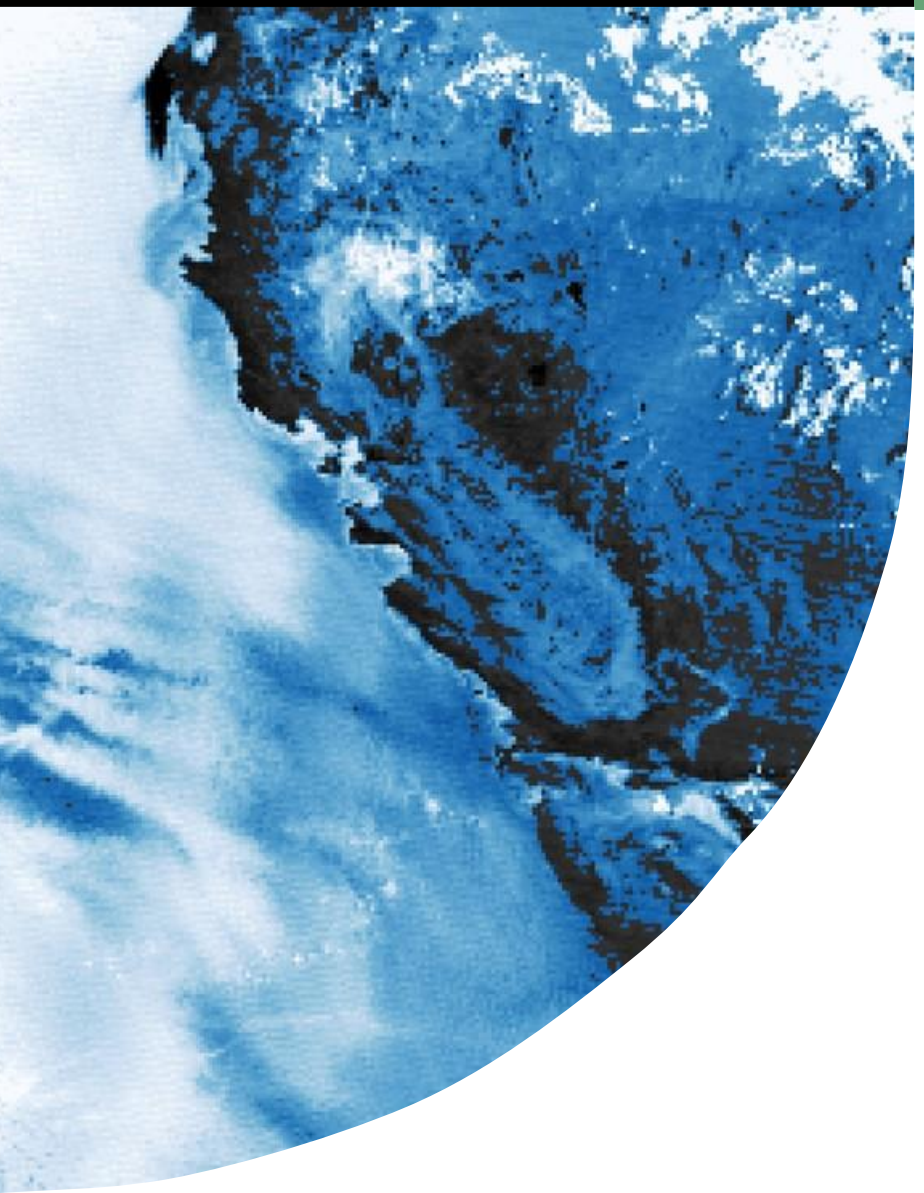
National Aeronautics and Space Administration



California Coast Ecological Conservation

Coupling Satellite Observations and Machine Learning to Understand Future Fog Prevalence Along the Coast Redwood Range

Macey Hartmann, Aubrey Palmer, Paul Seibert, & David Wilcox (Analytical Mechanics Associates)

California – Ames | Summer 2025

Background

Image credits: National Park Service, National Park Service, Tomwsulcer, Greg Schechter, National Park Service

Community Concerns

The future fog presence in the coast redwood range is uncertain.

Image credits: National Park Service, Thewellman, Marty Aligata

Project Partner

Save the Redwoods League

Image Credits: pellaea, National Parks Service

"The mission of Save the Redwoods League is to protect and restore redwood forests and connect people with their peace and beauty so these wonders of the natural world flourish."

- *Save the Redwoods League*

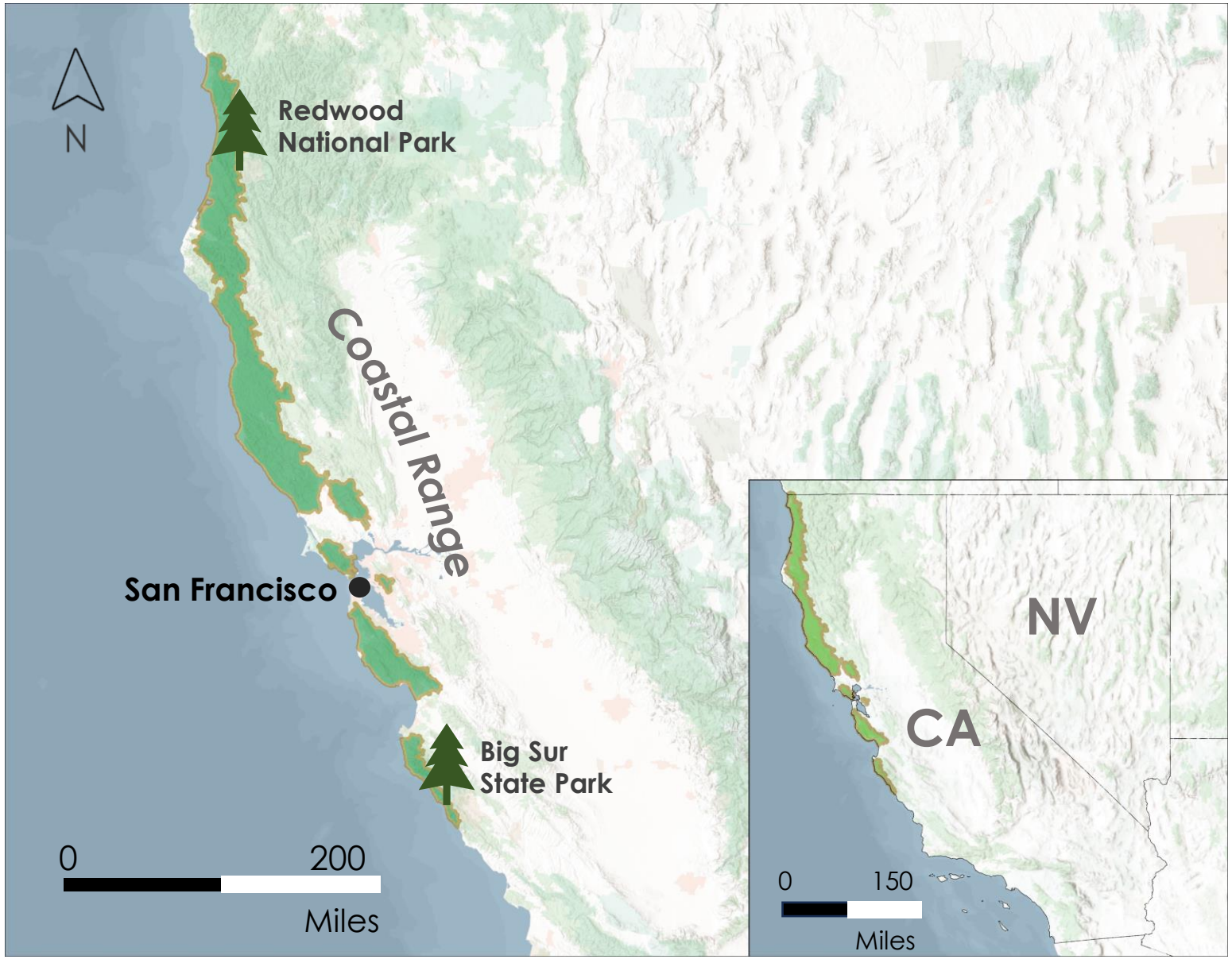
Study Area

Redwood Forests

Coast Redwood Range

Time period
1999–2025 (*historical*)
2080–2090 (*future*)

Fog season
June–September



Project Objectives

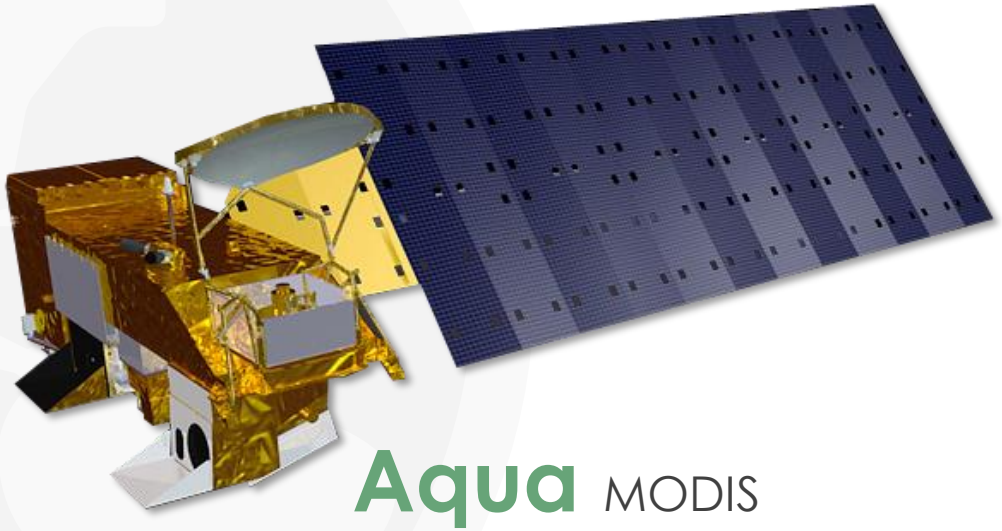
Build a historical fog model

Create projections of future fog to the year 2090

Analyze past, present, and future fog trends

Earth Observations

GOES-18 ABI
Geostationary Operational
Environmental Satellite
Advanced Baseline Imager



Aqua MODIS
Moderate-resolution
Imaging Spectroradiometer

Image Credit: NOAA/Lockheed Martin, NASA

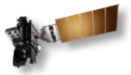
Methods

1999 | 2025 | 2080

Past Fog

(Torregrosa et al., 2015) + Team's Reconstruction

from Satellite Data (GOES)



Climate Projection Data (CMIP)

Machine Learning

Future Fog

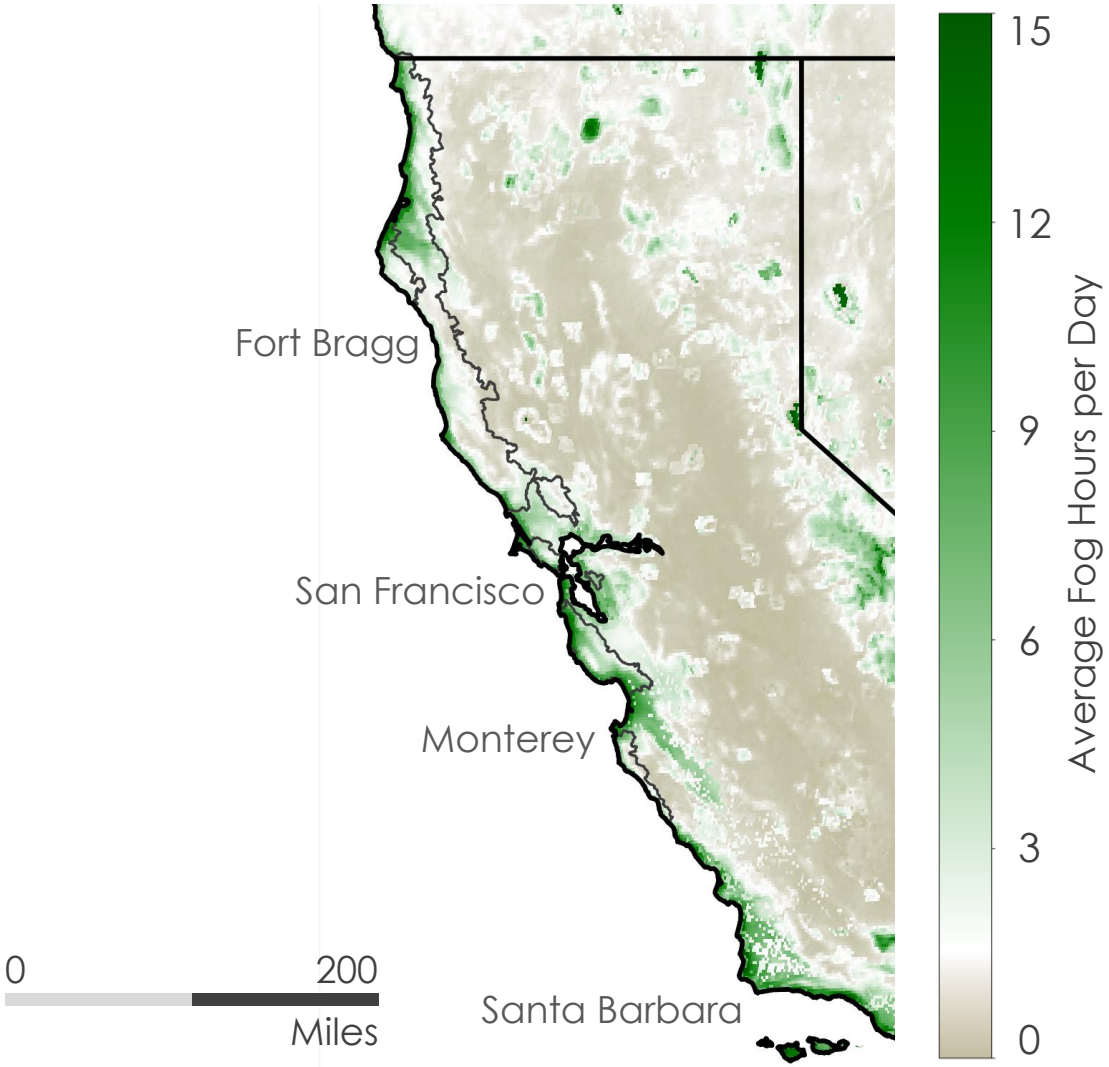
Fog Projections

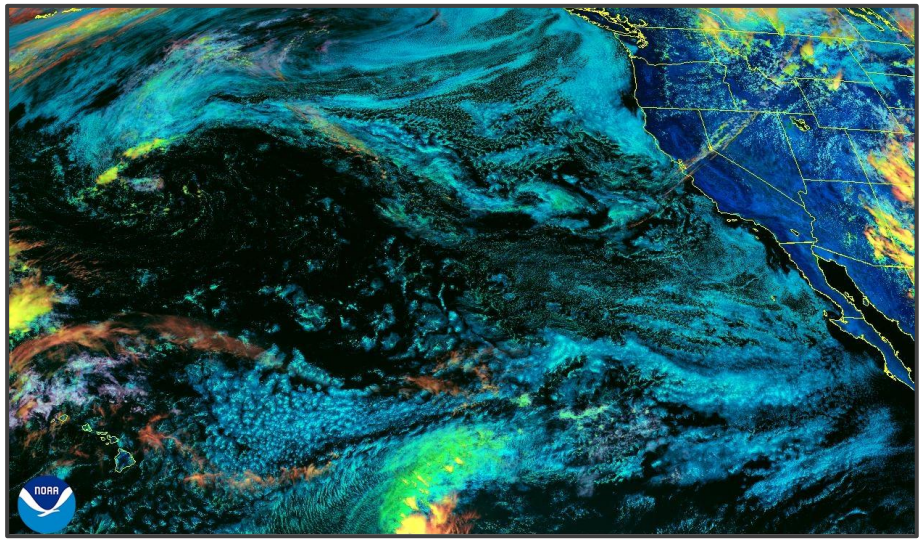
Used for validation/further research:

- Weather station data (CIMIS)
- Sea & land surface temperature (Aqua MODIS)

Image Credit: NOAA/Lockheed Martin

Historical Hours of Fog





GOES Cloud Top Phase

Image Credit: NOAA

Cloud Top Phase

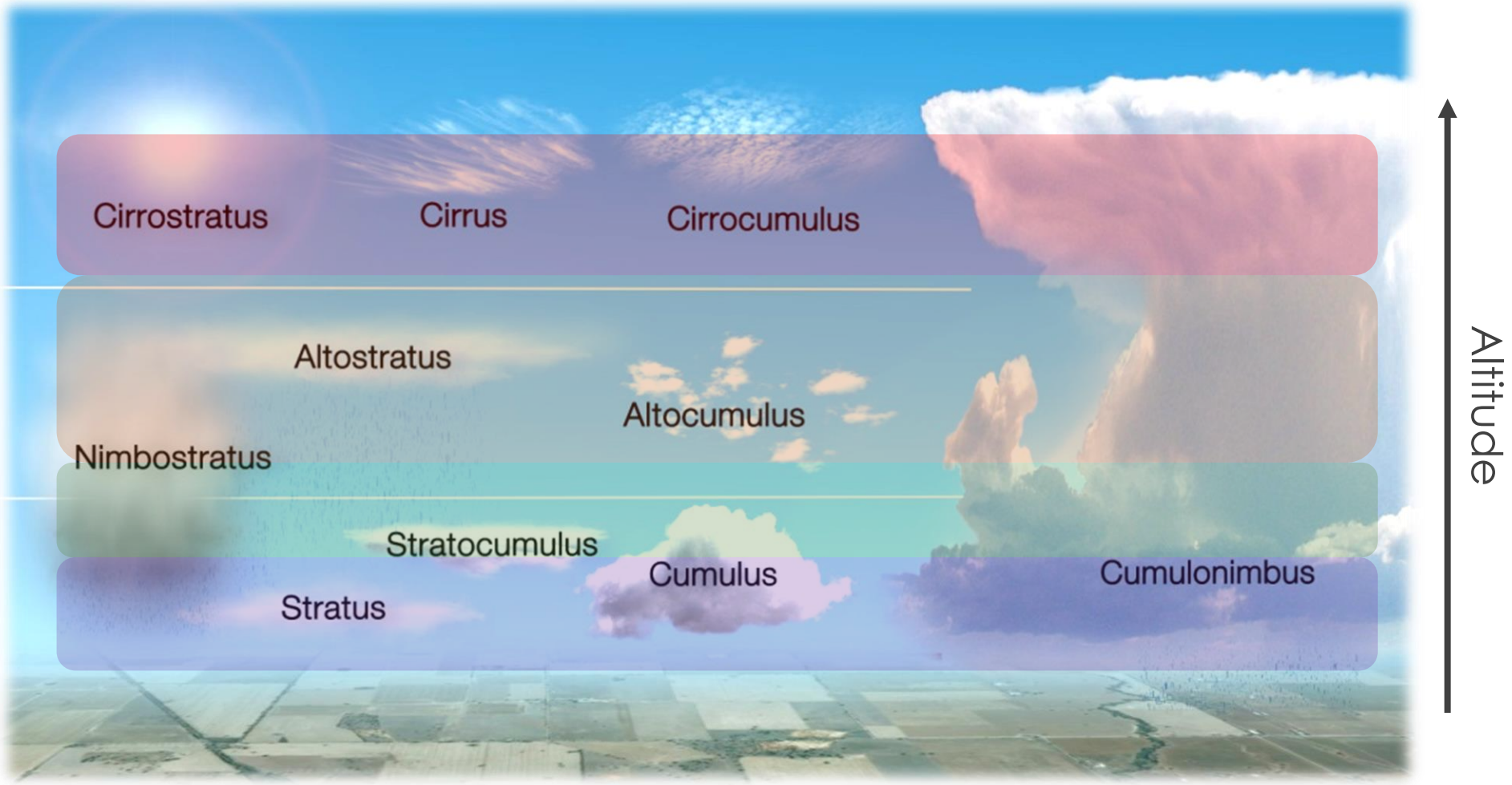
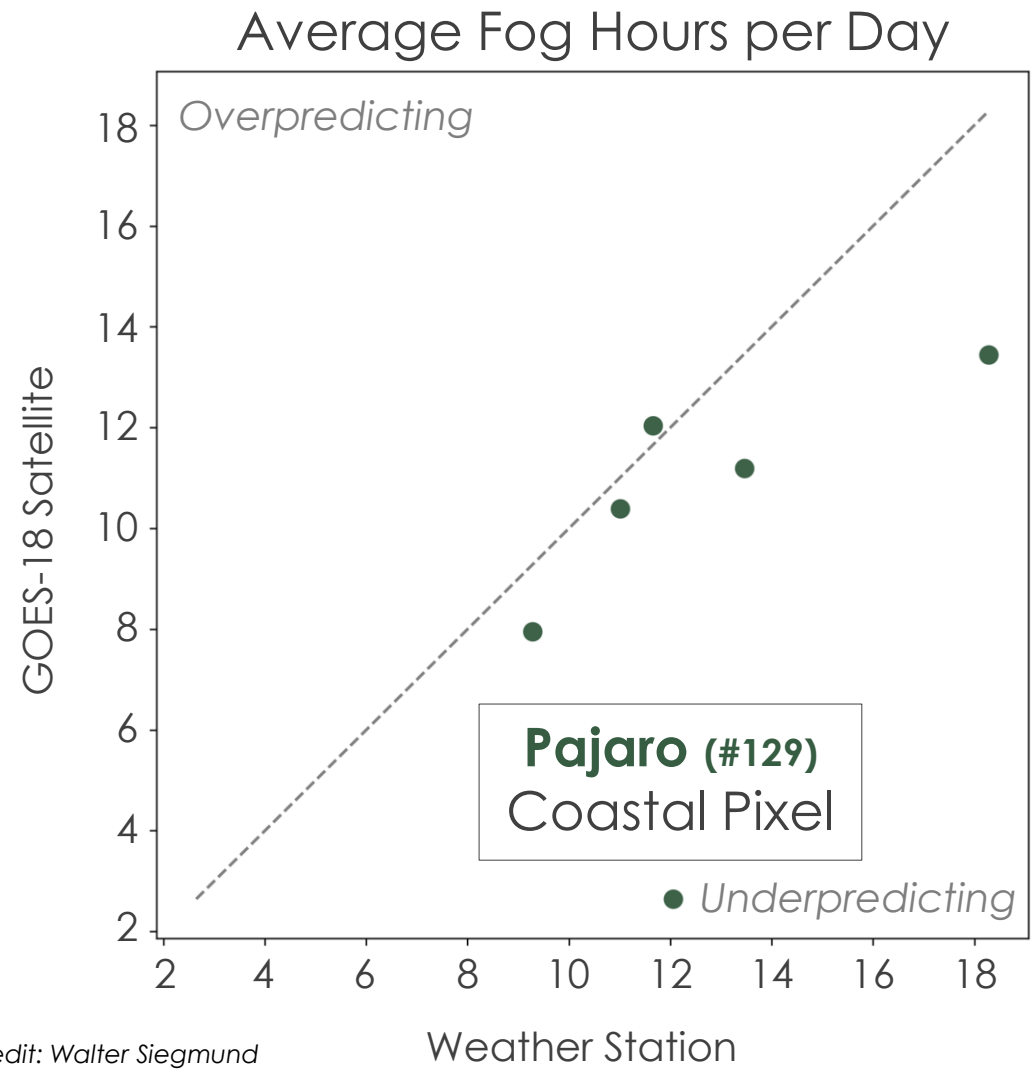


Image Credit: Australian Bureau of Meteorology

Is It Accurate?

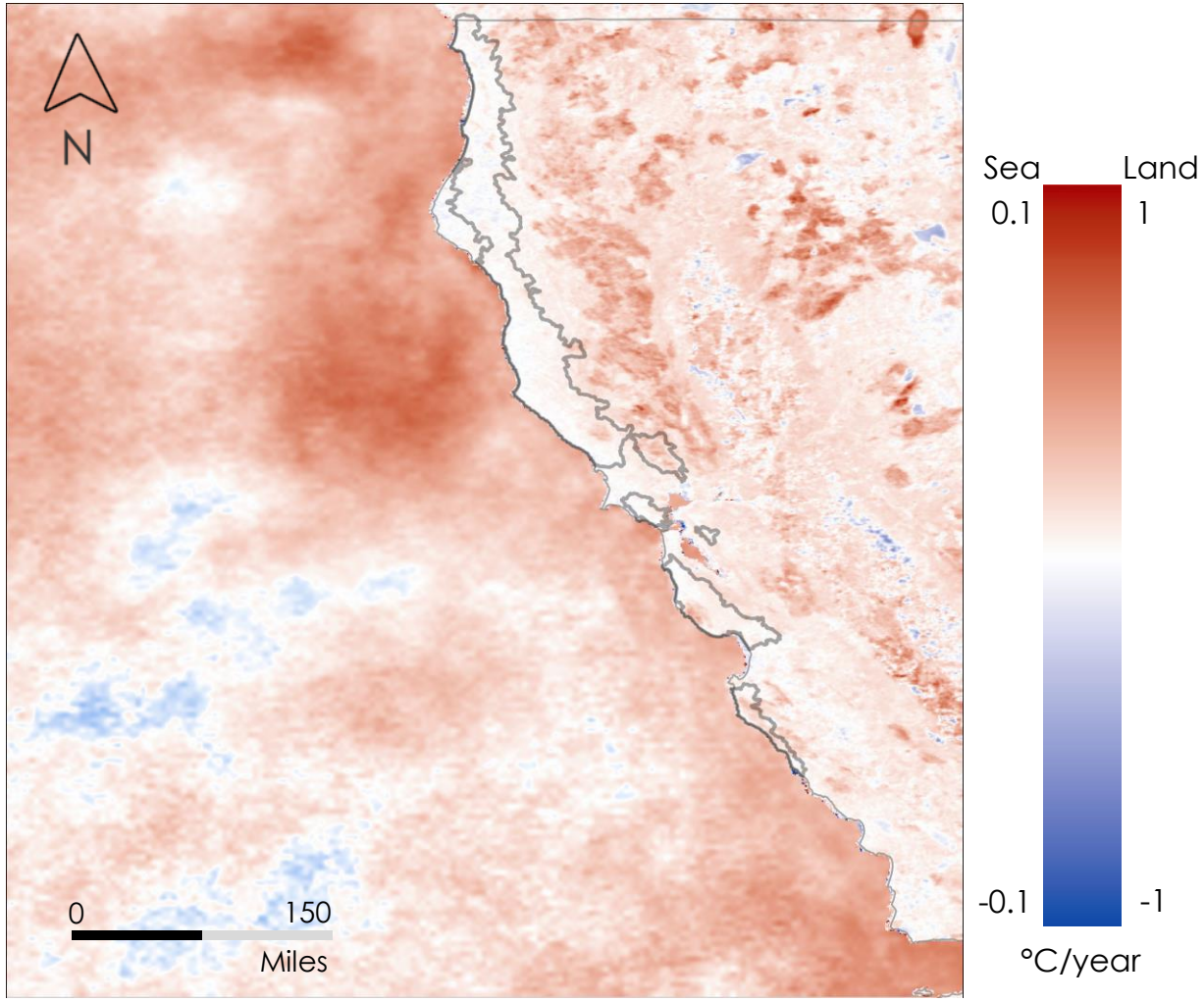


Pacific Grove (#193)
Likely Sensor Error

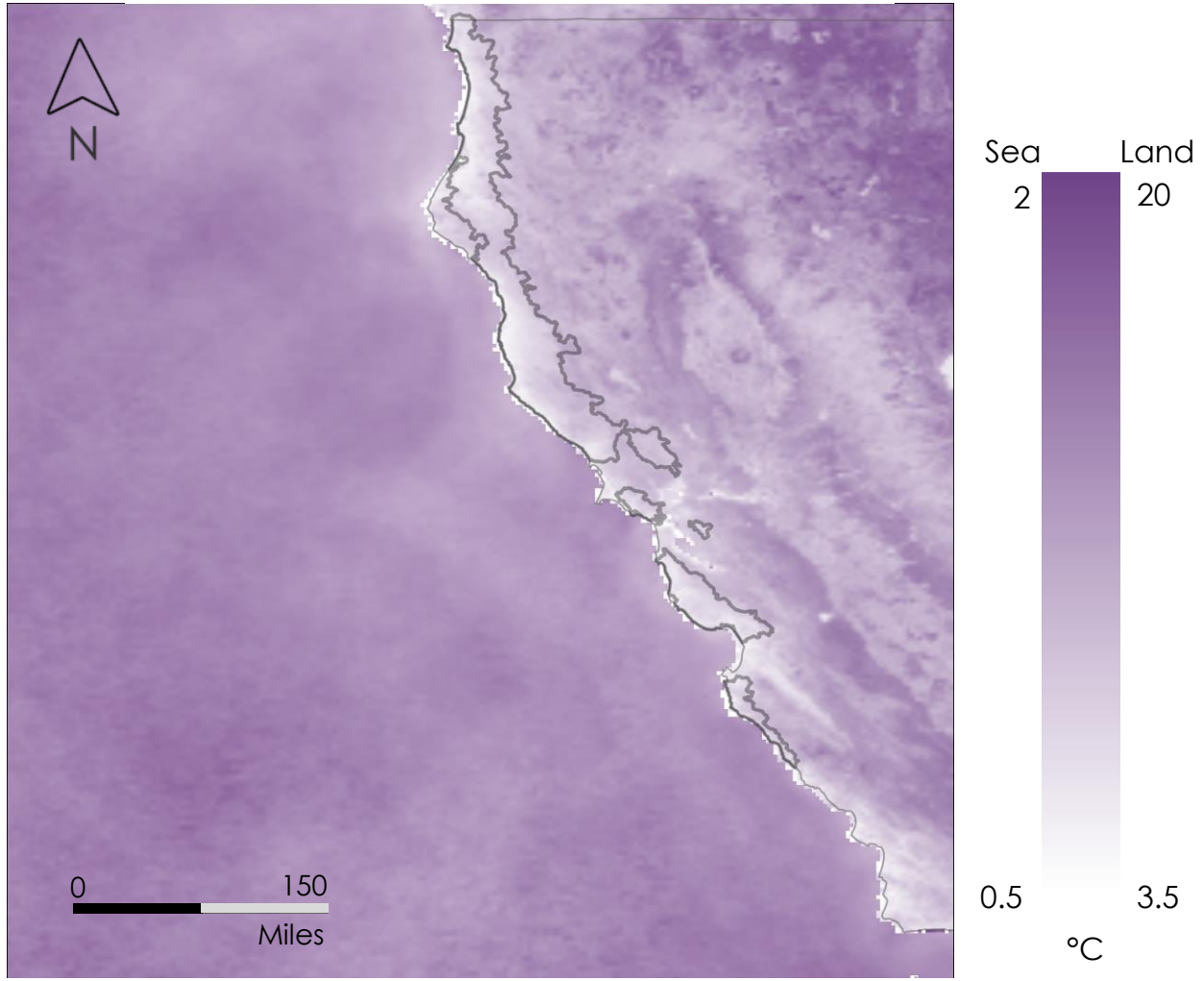
Image credit: Walter Siegmund

Aqua MODIS – Land & Sea Skin Temperature

Average Temp. Trend (2002–2025)



Standard Deviation (2002–2025)

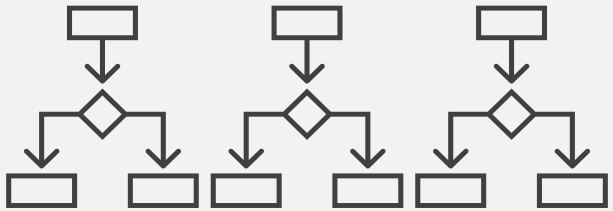


Random Forest Development

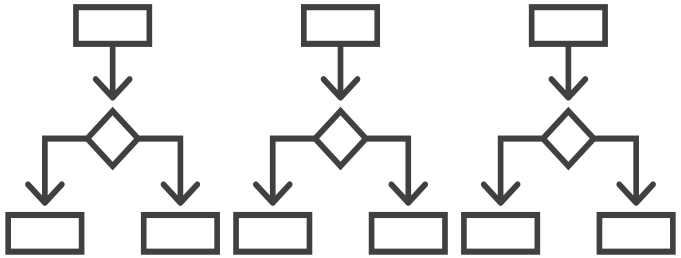
Train

Past climate data

Past fog/low cloud cover



Test

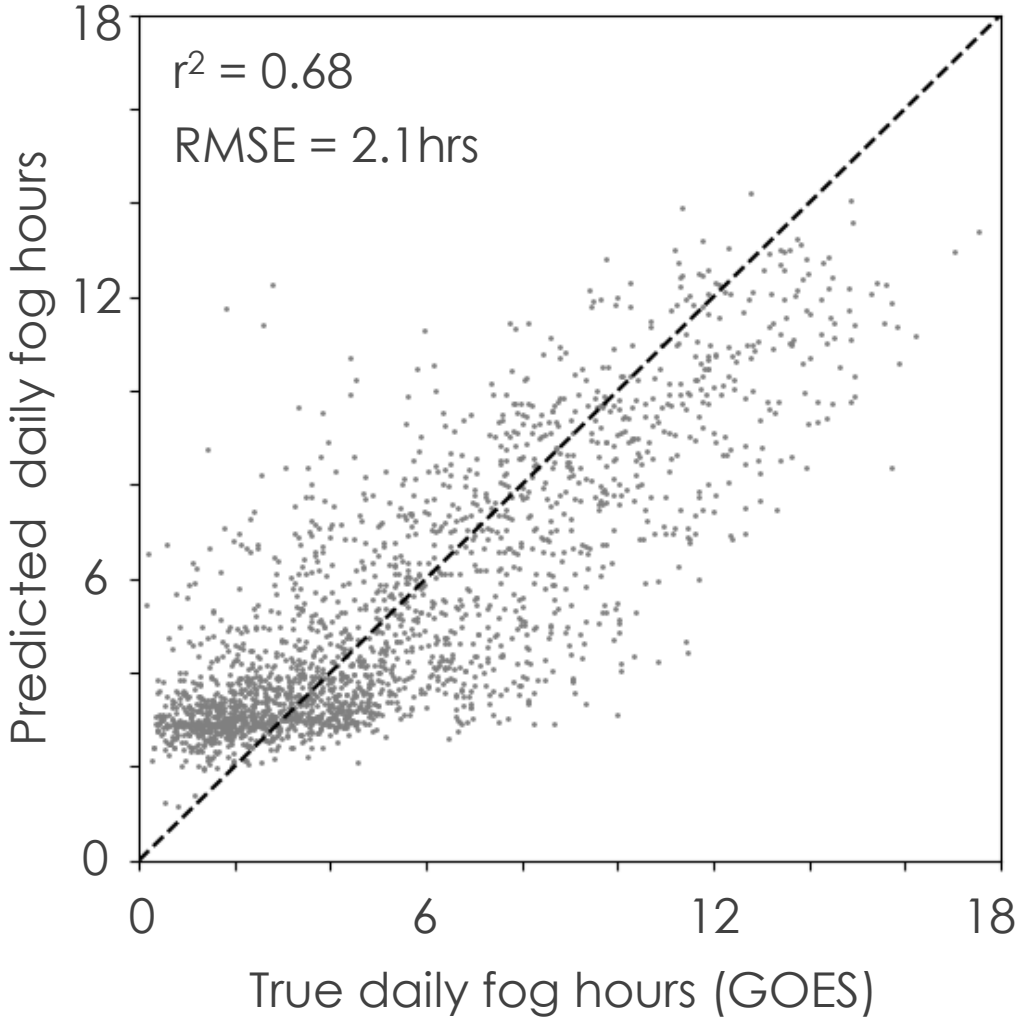


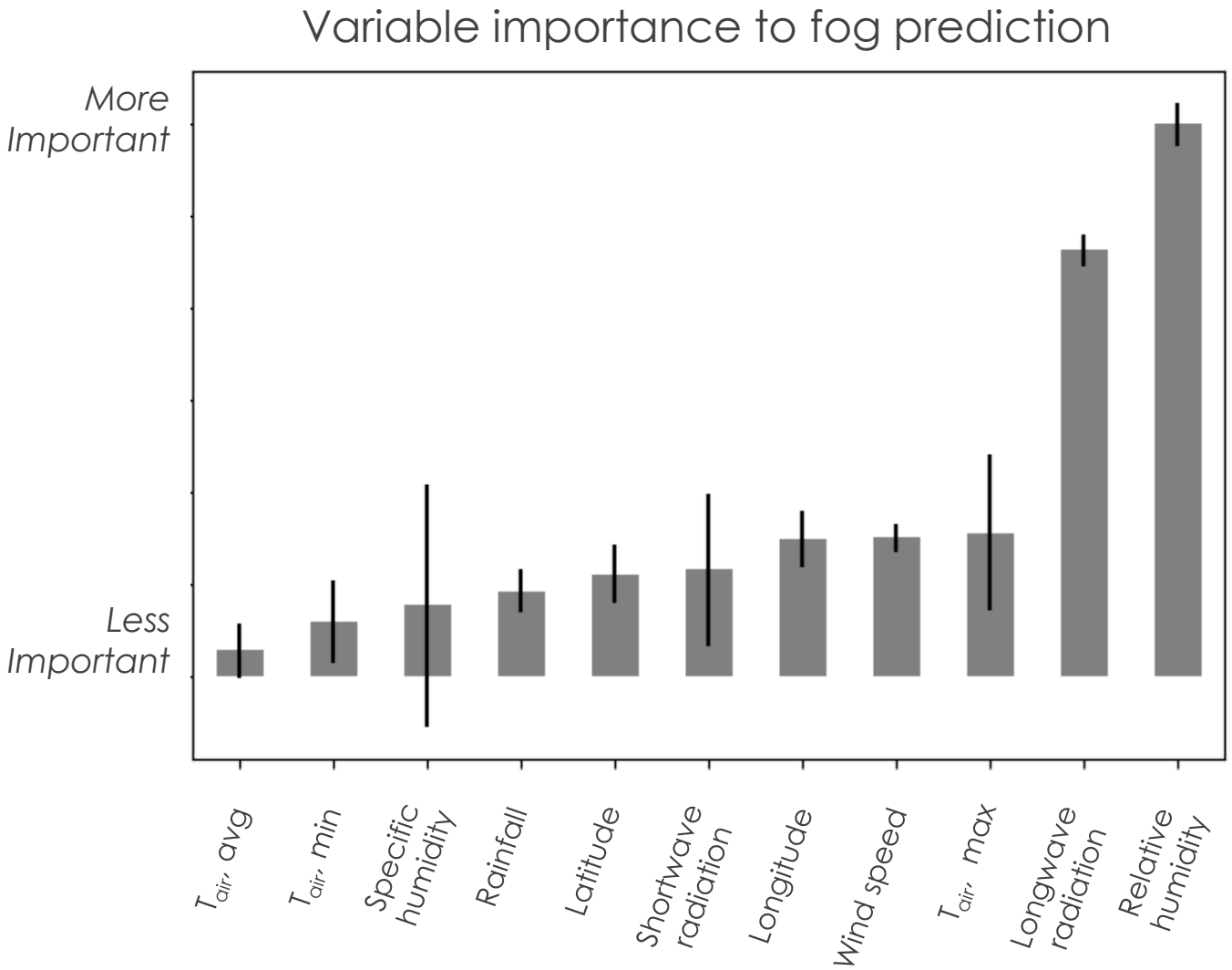
Predicted fog/low cloud cover

Tune

Tune hyperparameters and **repeat procedure**

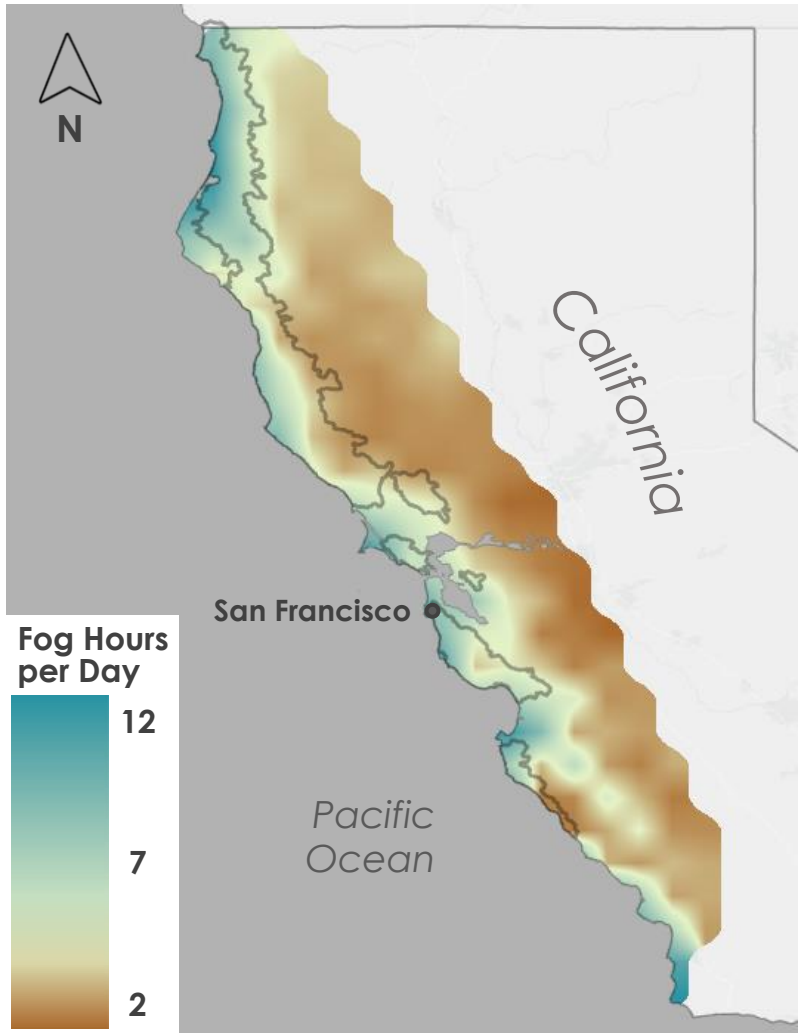
Random Forest Analysis



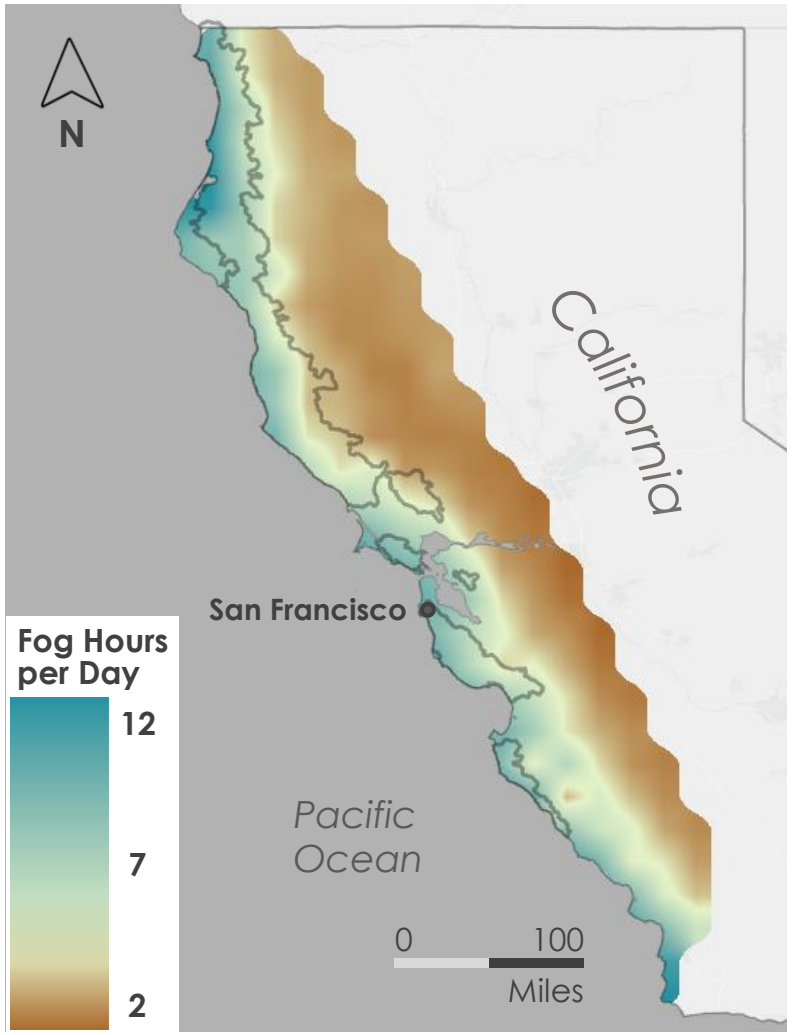


Future Fog Increasing

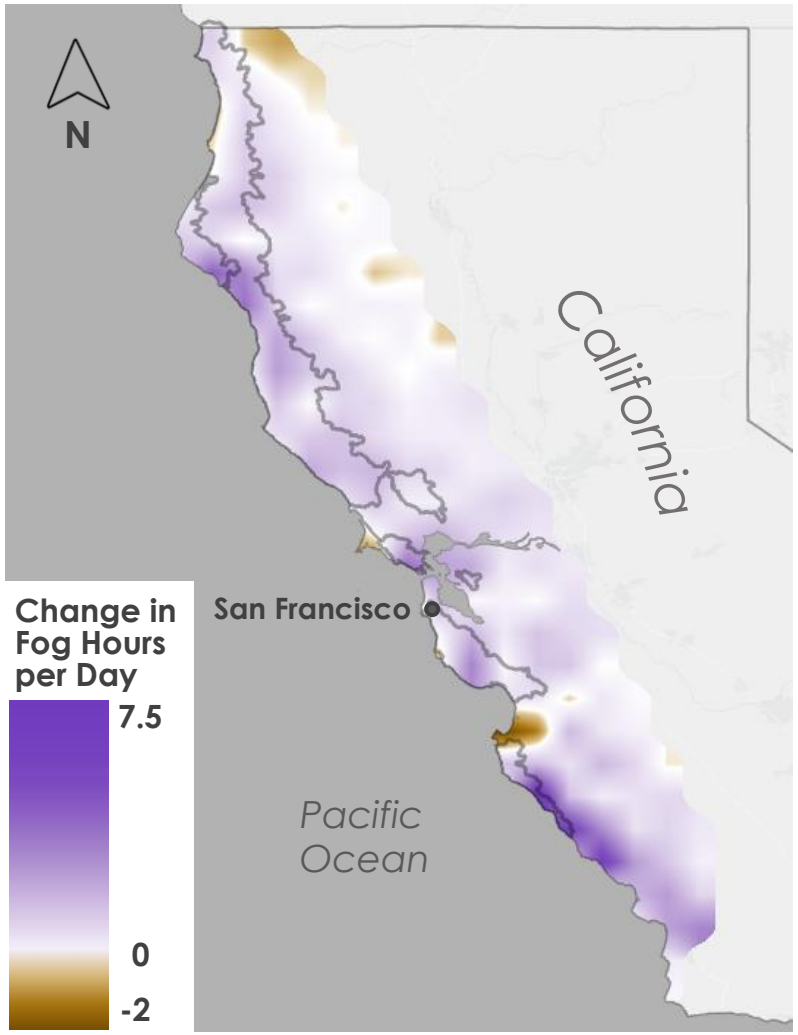
Past Fog (1999–2009)



Future Fog (2080–2090)



Change in Fog Prevalence



Limitations

Satellite and Sensor ranges & products

Model training data range & certainty

Coarse Resolution of final product

Feasibility & Partner Implementation

Remote Sensing is an effective way to study fog

Data Acquisition is **time-consuming** and **scattered**

Difficult to make conservation decisions with **low-confidence, low-resolution** data

Overall methodology is highly effective

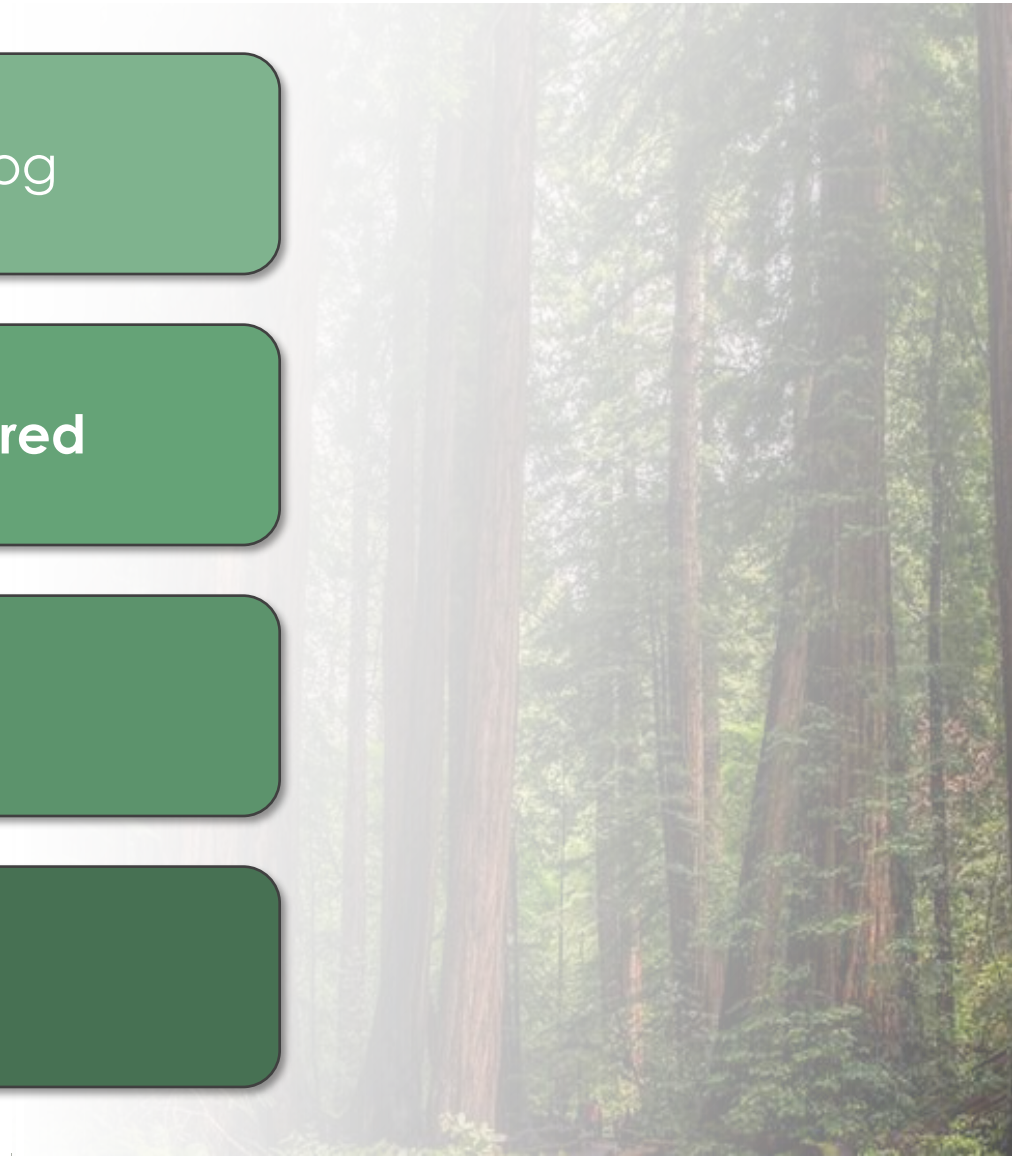
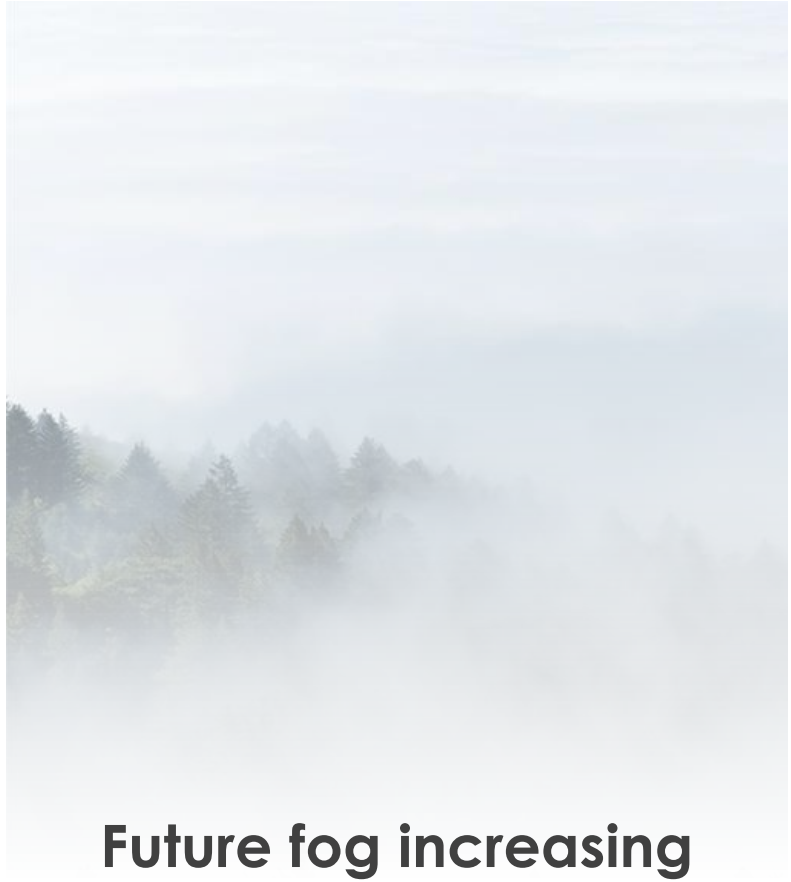


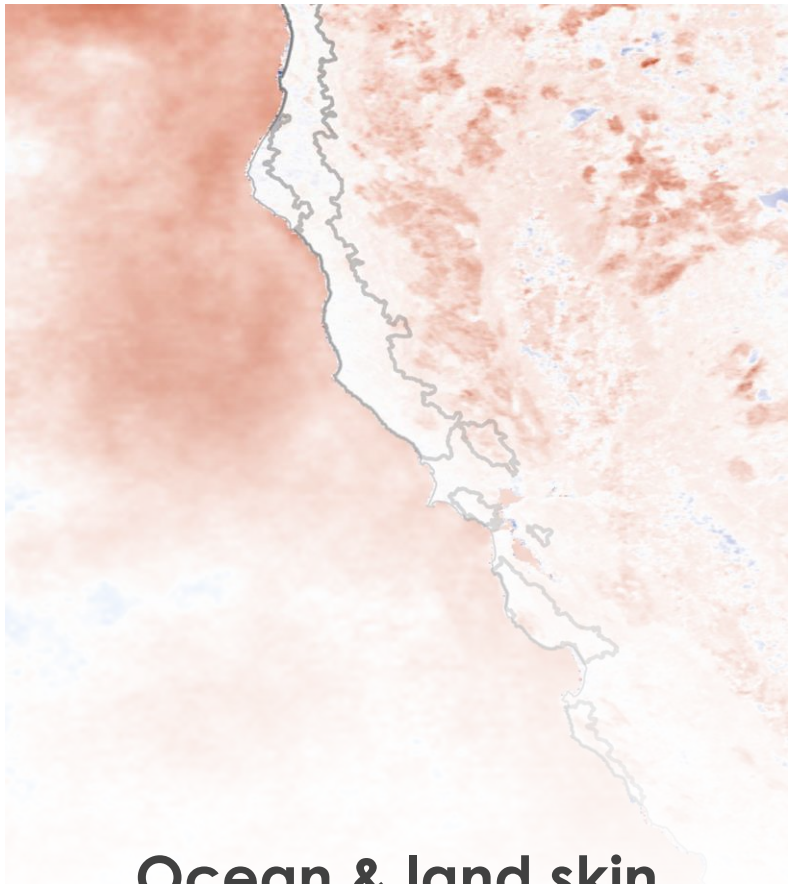
Image credit: Justin Brown

Conclusions

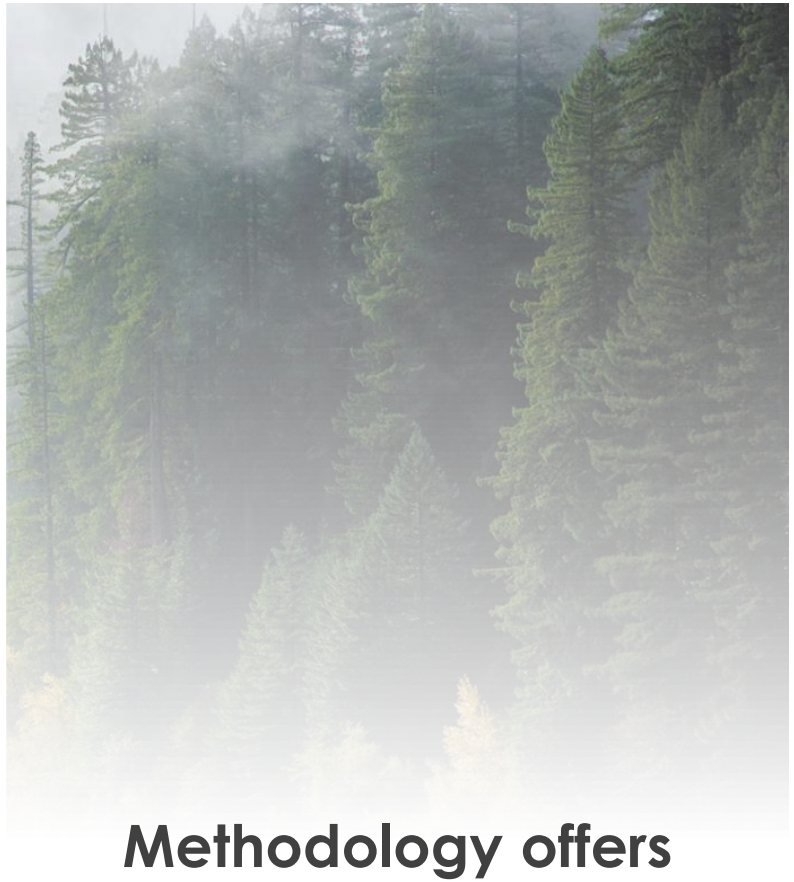
Image credit: Corey Ageopian, David Baselt



Future fog increasing across nearly the entire coast redwood range



Ocean & land skin temps increasing



Methodology offers insight into past & future fog trends

Acknowledgements

Advisors

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- Laura Lalemand (Save the Redwoods League – Senior Scientist)

Center Lead & Fellow

- Katie Miller (NASA DEVELOP – ARC)
- Maya Hall (NASA DEVELOP – ARC)